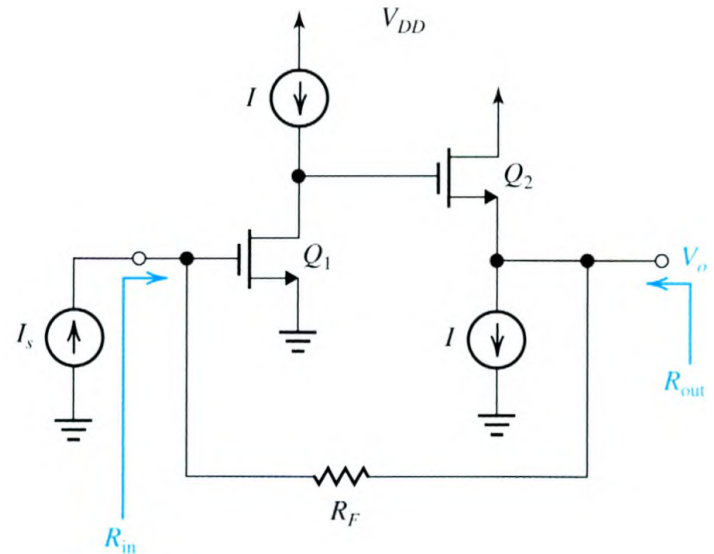


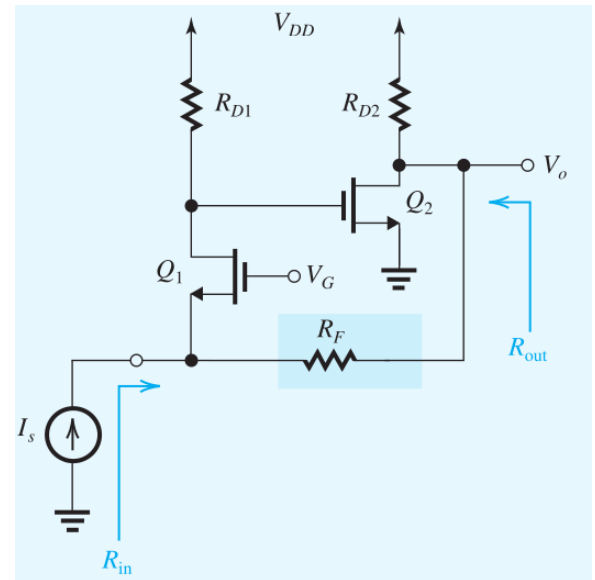
Question 1

The feedback transresistance amplifier in Figure below utilizes two identical MOSFETs biased by ideal current sources $I = 0.3 \text{ mA}$. The MOSFETs are sized to operate at $V_{ov} = 0.15 \text{ V}$ and have $V_t = 0.35 \text{ V}$ and $V_A = 6 \text{ V}$. The feedback resistance $R_F = 20 \text{ k}\Omega$.

- If I_s has a zero DC component, find the DC voltage at the input, at the drain of Q_1 , and at the output.
- Find g_m and r_o of Q_1 and Q_2 .
- Provide the A circuit and derive an expression for A in terms of g_{m1} , r_{o1} , g_{m2} , r_{o2} , and R_F .
- What is β ? Give an expression for the loop gain $A\beta$ and the amount of feedback $(1 + A\beta)$.
- Derive an expression for A_f .
- Derive expressions for R_i , R_{in} , R_o , and R_{out} .
- Evaluate A, β , $A\beta$, A_f , R_i , R_o , R_{in} , and R_{out} for the component values given.

**Question 2**

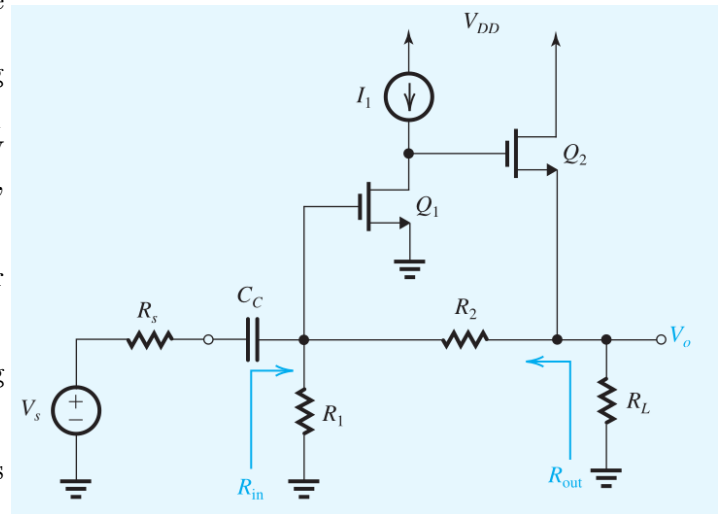
For the feedback amplifier in figure below, select a value for R_F that results in an ideal closed-loop gain $A_f \equiv v_o/I_s = -20 \text{ V/mA}$. Then, analyze the circuit to determine the actual value of A_f realized. As well, determine R_{in} and R_{out} . Transistors Q_1 and Q_2 are operated so that $g_{m1} = g_{m2} = 2 \text{ mA/V}$ and r_{o1} and r_{o2} can be neglected. Also, $R_{D1} = R_{D2} = 20 \text{ k}\Omega$.



Question 3

Figure below shows a shunt–shunt feedback amplifier. The MOSFETs have $V_{tn} = 0.6$ V, $V_A = 20$ V, and $\mu_n C_{ox} = 200 \mu\text{A}/\text{V}^2$. The power supply $V_{DD} = 3.3$ V, and $R_L = 2$ k Ω . The coupling capacitor C_C can be assumed to be very large.

- Perform a DC design to meet the following specifications: $I_{D1} = 100 \mu\text{A}$, $I_{D2} = 1$ mA, $I_{R2}, R_1 = 10 \mu\text{A}$, $V_{OV1} = V_{OV2} = 0.2$ V. Neglect the Early effect. Specify the values required for I_1 , R_1 , R_2 , $(W/L)_1$, and $(W/L)_2$.
- Find an expression for β and hence an expression for the ideal value of V_o/V_s .
- Find the value of R_s that results in V_o/V_s being ideally -6 V/V.
- Find the A circuit and use it to determine the values of A, R_i , and R_o .
- Find the value obtained for V_o/V_s .
- Find R_{in} and R_{out} .

**Question 4**

The feedback current amplifier in figure (a) can be thought of as a “super” CG transistor. Note that rather than connecting the gate of Q_2 to signal ground, an amplifier is placed between source and gate.

- If μ is very large, what is the signal voltage at the input terminal? What is the input resistance? What is the current gain I_o/I_s ?
- For finite μ but assuming that the input resistance of the amplifier μ is very large, find the A circuit and derive expressions for A, R_i , and R_o .
- What is the value of β ?
- Find $A\beta$ and A_f . If μ is large, what is the value of A_f ?
- Find R_{in} and R_{out} assuming the loop gain is large.
- The “super” CG transistor can be utilized in the cascode configuration shown in figure (b), where V_G is a DC bias voltage. Replacing Q_1 by its small-signal model, use the analogy of the resulting circuit to that in figure (a) to find I_o and R_{out} .

